

College of Industrial Technology
King Mongkut's University of Technology North Bangkok



Midterm Examination of Semester 1

Year: 2013

Subject: 392131 Physics I

Section: 15 - 16

Date: 31 July 2013

Time: 08.00 – 10.00

Name: _____ ID: _____ Class: _____

Instructions:

1. Cheating will result in failure of all classes registered for the current semester. Students who are caught cheating will also be denied registering for the following semester.
 2. No documents are allowed to be taken inside the examination room.
 3. Textbooks are **NOT** allowed.
 4. Express your answer in English. Other languages will be discarded.
 5. A scientific calculator is allowed.
 6. Electronic communication devices are **NOT** allowed during the examination.
 7. The examination has 7 pages (including this page), 24 questions and a total score of 70 points.
 8. Write all your answers on this exam sheet.
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1. Show that the expression $s = ut + \frac{1}{2}at^2$ is dimensionally correct (1 point)

2. Kinetic energy KE has dimensions $\text{kg} \cdot \text{m}^2/\text{s}^2$. It can be written in terms of the momentum p and mass m as (1 point)

$$KE = \frac{p^2}{2m}$$

Determine the proper units for momentum using dimensional analysis.

3. How many significant figures are there in (3 points)

a) 01.123450

b) 4.708×10^9

c) 0.04080

d) 0.0053

e) 1.389×10^{-6}

f) 42.44 ± 0.01

4. Use the rules for significant figures to find the answer to the addition problem. (1 point)

$$0.243 + 15.3 + 110.17 + 4.321 = \dots\dots\dots$$

5. A farmer measures the distance around a rectangular field. The length of the long sides of the rectangle is found to be 38.44 m, and the length of the short sides is found to be 19.5 m. What is the total distance around the field? (1 point)

6. Multiply 2.079×10^2 m by 0.082×10^{-1} m taking in to account significant figures. (2 points)

7. What is the area of a circle of radius 3.8×10^4 cm? (1 point)

Use this information to answer the question below

$$1 \text{ mi} = 1609 \text{ m} = 1.609 \text{ km}$$

$$1 \text{ ft} = 0.3048 \text{ m} = 30.48 \text{ cm}$$

$$1 \text{ m} = 39.37 \text{ in.} = 3.281 \text{ ft}$$

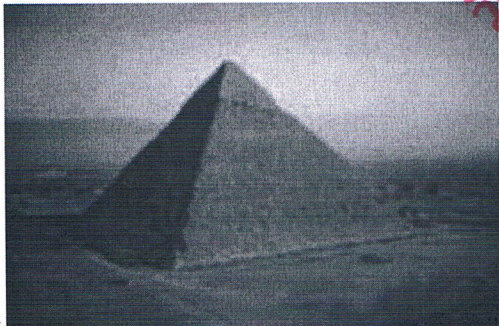
$$1 \text{ in.} = 0.0254 \text{ m} = 2.54 \text{ cm}$$

$$1 \text{ gallon} = 3.786 \text{ liter}$$

$$1 \text{ fathom} = 6 \text{ ft}$$

8. Convert $4.80 \times 10^3 \text{ kg/cm}^3$ to g/m^3 . (2 points)

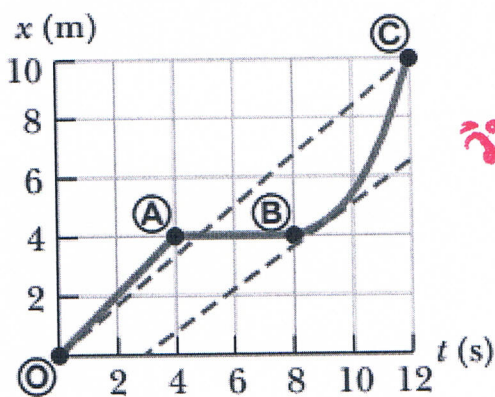
9. The volume of a wallet is 8.50 in^3 . Convert this value to m^3 . (2 points)



10. The base of a pyramid covers an area of 13.0 acres (1 acre = $43,560 \text{ ft}^2$) and has a height of 481 ft. If the volume of a pyramid is given by the expression $V = \frac{bh}{3}$, where b is the area of the base and h is the height, Find the volume of this pyramid in cubic meters. (4 points)

11. Find the polar coordinates corresponding to $(x, y) = (-4 \text{ m}, -5 \text{ m})$. (2 points)

12. Two points are given in polar coordinates by $(x,y) = (-5.00 \text{ m}, -3 \text{ m})$ and, $(r, \theta) = (7.00 \text{ m}, 150.0^\circ)$ respectively. What is the distance between them? (4 points)



13. A train moves slowly along a straight portion of track according to the graph of position versus time in Figure. Find
 (a) the average velocity for the total trip,
 (b) the average velocity during the first 8.00 s of motion,
 (c) the instantaneous velocity at $t = 10 \text{ s}$. (4 points)

14. A particle moves according to the equation $x = 10t^2$ where x is in meters and t is in seconds. Find the average velocity for the time interval from 2.00s to 3.00s. (3 points)

15. A motorist drives north for 30.0 minutes at 90.0 km/h and then stops for 20.0 minutes. He then continues north, traveling 300 km in 4.00 h. (4 points)

- (a) What is his total displacement?
- (b) What is his average velocity?

16. A truck covers 50.0 m in 7.0 s while uniformly slowing down to a final velocity of 2.80 m/s. (4 points)

- (a) Find the truck's original speed.
- (b) Find its acceleration.

17. A speedboat moving at 30.0 m/s approaches a no-wake buoy marker 100 m ahead. The pilot slows the boat with a constant acceleration of -3.50 m/s^2 by reducing the throttle. (4 points)

- (a) How long does it take for the boat to reach the buoy?
- (b) What is the velocity of the boat when it reaches the buoy?

18. The driver of a car slams on the brakes when he sees a tree blocking the road. The car slows uniformly with an acceleration of -5.60 m/s^2 for 4.20s, making straight skid marks 62.4 m long ending at the tree. With what speed does the car then strike the tree? (2 points)

19. A ball starts from rest and accelerates at 0.500 m/s^2 while moving down an inclined plane 9.00 m long. When it reaches the bottom, the ball rolls up another plane, where, after moving 15.0 m, it comes to rest. (4 points)

(a) What is the speed of the ball at the bottom of the first plane?

(b) How long does it take to roll down the first plane?

20. A ball is thrown vertically upward with a speed of 300 m/s. (4 points)

(a) How high does it rise?

(b) How long does it take to reach its highest point?

21. A small mailbag is released from a helicopter that is rising steadily at 1.50 m/s. After 2.00 s, (5 points)

(a) what is the speed of the mailbag?

(b) how far is it below the helicopter?

22. A woman is reported to have fallen 144 ft from the 17th floor of a building, landing on a metal ventilator box, which she crushed to a depth of 18.0 in. She suffered only minor injuries. Neglecting air resistance, calculate (4 points)

(a) the speed of the woman just before she collided with the ventilator.

(b) her average acceleration while in contact with the box.

23. A freely falling object requires 1.50 s to travel the last 30.0 m before it hits the ground. From what height above the ground did it fall? (hints : at the start point initial velocity is zero) (4 points)

24. A rocket moves straight upward, starting from rest with an acceleration of 30 m/s^2 . It runs out of fuel at the end of 5.00 s and continues to coast upward, reaching a maximum height before falling back to Earth. Find the maximum height the rocket reaches. (4 points)

Good Luck

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